

CSS Fundamentals – Complete Notes (5–6 Pages)

1. Introduction to CSS

What is CSS?

CSS (Cascading Style Sheets) is a stylesheet language used to control the appearance and layout of web pages.

HTML defines the structure of a webpage, while CSS defines how it looks.

Benefits of CSS

- Separates content from design
- Improves website maintainability
- Enables responsive layouts
- Reduces code duplication
- Improves user experience

Basic Syntax

```
selector {  
  property: value;  
}
```

Example

```
<h1>Welcome to CSS</h1>
```

```
h1 {  
  color: blue;  
  font-size: 32px;  
}
```

2. CSS Selectors

Selectors are used to target HTML elements.

Element Selector

```
p {  
    color: red;  
}
```

Targets all paragraph elements.

Class Selector

```
.highlight {  
    background-color: yellow;  
}
```

```
<p class="highlight">Important Text</p>
```

ID Selector

```
#header {  
    background-color: black;  
    color: white;  
}
```

```
<div id="header">Header</div>
```

Universal Selector

```
* {  
    margin: 0;  
    padding: 0;  
}
```

Targets all elements.

Pseudo-Class

```
button:hover {  
    background-color: green;  
}
```

Applies styles when hovering over a button.

3. Colors, Backgrounds, and Text

Colors

Color Name

```
color: red;
```

HEX Value

```
color: #ff0000;
```

RGB

```
color: rgb(255, 0, 0);
```

Background Color

```
background-color: lightblue;
```

Background Image

```
background-image: url("image.jpg");  
background-size: cover;
```

Text Formatting

Text Alignment

```
text-align: center;
```

Text Decoration

```
text-decoration: underline;
```

Text Transform

```
text-transform: uppercase;
```

Text Shadow

```
text-shadow: 2px 2px 4px gray;
```

Example:

```
h1 {  
  text-align: center;  
  text-transform: uppercase;  
}
```

4. Fonts, Borders, Margins, and Padding

Fonts

Font Family

```
font-family: Arial, sans-serif;
```

Font Size

```
font-size: 20px;
```

Font Weight

```
font-weight: bold;
```

Font Style

```
font-style: italic;
```

Borders

Basic Border

```
border: 2px solid black;
```

Border Radius

```
border-radius: 10px;
```

Individual Sides

```
border-top: 2px solid red;  
border-bottom: 2px solid blue;
```

Example:

```
.box {  
  border: 2px solid black;  
  border-radius: 8px;  
}
```

Margins

Margin creates space outside an element.

```
margin: 20px;
```

```
margin: 10px 20px;
```

Padding

Padding creates space inside an element.

```
padding: 20px;
```

Example:

```
.card {  
  border: 1px solid gray;  
  padding: 15px;  
  margin: 20px;  
}
```

5. CSS Box Model and Positioning

CSS Box Model

Every HTML element consists of:

1. Content
2. Padding
3. Border
4. Margin

Visualization:

```
Margin  
└─ Border  
   └─ Padding  
      └─ Content
```

Example:

```
.box {  
  width: 200px;  
  padding: 20px;  
  border: 5px solid black;  
  margin: 30px;  
}
```

Width and Height

```
width: 300px;  
height: 200px;
```

```
min-width: 200px;  
max-width: 800px;
```

Positioning

Static

Default position.

```
position: static;
```

Relative

Moves relative to its original position.

```
position: relative;  
left: 20px;
```

Absolute

Positions relative to nearest positioned parent.

```
position: absolute;  
top: 10px;  
right: 10px;
```

Fixed

Stays fixed on screen.

```
position: fixed;  
bottom: 10px;  
right: 10px;
```

Sticky

Acts as relative until scrolling.

```
position: sticky;  
top: 0;
```

6. Flexbox and Grid Layout

Flexbox

Used for one-dimensional layouts.

Flex Container

```
.container {  
  display: flex;  
}
```

Main Properties

```
justify-content: center;  
align-items: center;  
flex-direction: row;
```


Example

```
.container {  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
}
```

```
<div class="container">  
  <div>Item 1</div>  
  <div>Item 2</div>  
</div>
```

CSS Grid

Used for two-dimensional layouts.

Grid Container

```
.container {  
  display: grid;  
  grid-template-columns: repeat(3, 1fr);  
  gap: 20px;  
}
```

Example

```
<div class="container">  
  <div>1</div>  
  <div>2</div>  
  <div>3</div>  
</div>
```

7. Transforms, Transitions, and Animations

Transform

Rotate

```
transform: rotate(45deg);
```

Scale

```
transform: scale(1.2);
```

Translate

```
transform: translateX(50px);
```

Transition

Creates smooth effects.

```
button {  
  transition: all 0.3s ease;  
}
```

```
button:hover {  
  background-color: blue;  
}
```

Animation

Define Animation

```
@keyframes move {  
  from {  
    left: 0;  
  }  
}
```

```
    to {  
        left: 200px;  
    }  
}
```

Apply Animation

```
.box {  
    position: relative;  
    animation: move 2s infinite;  
}
```

8. Responsive Design

Responsive design ensures websites work on all screen sizes.

Media Query

```
@media (max-width: 768px) {  
    body {  
        background-color: lightgray;  
    }  
}
```

Example

```
.container {  
    display: flex;  
}  
  
@media (max-width: 768px) {  
    .container {  
        flex-direction: column;  
    }  
}
```

9. CSS Best Practices

1. Use external CSS files.
 2. Prefer classes over IDs for styling.
 3. Use meaningful class names.
 4. Avoid excessive nesting.
 5. Use Flexbox and Grid for layouts.
 6. Use CSS variables for consistency.
 7. Keep styles modular and reusable.
 8. Optimize for mobile devices.
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Mini Project Example

HTML

```
<div class="card">
  <h2>CSS Notes</h2>
  <p>Learning CSS Fundamentals</p>
</div>
```

CSS

```
.card {
  width: 300px;
  margin: 20px auto;
  padding: 20px;

  border: 1px solid #ddd;
  border-radius: 10px;

  box-shadow: 0 4px 10px rgba(0,0,0,0.2);

  transition: 0.3s;
}

.card:hover {
  transform: translateY(-5px);
}
```

Summary

CSS is used to style web pages and consists of:

- Selectors
- Colors and Backgrounds
- Text and Fonts
- Borders, Margins, Padding
- Box Model
- Positioning
- Flexbox
- Grid
- Transformations
- Transitions
- Animations
- Responsive Design

Mastering these concepts provides a strong foundation for modern web development.